

Real-World Robotics Datasets for Incremental Learning and Lifelong Learning

1. OpenLORIS-Object (Lifelong Robotic Vision Object Recognition)

Overview

OpenLORIS-Object is a robotic vision dataset designed to evaluate **lifelong and incremental learning algorithms in real robotic settings**. It was collected with **RGB-D sensors mounted on mobile robots** operating in indoor environments (e.g., homes and offices), capturing variations in **illumination, occlusion, object scale, and clutter**. ([Lifelong Robotic Vision](#))

Contents

- RGB and Depth images of everyday objects
- ~69 object instances across 19 categories
- Multiple environmental factors (illumination, occlusion, pixel size, clutter)
- Temporal image sequences with environmental variance
- ~215,000 total images for benchmarking incremental tasks ([Lifelong Robotic Vision](#))

Why It's Useful

- Real robot perception challenges (not synthetic)
- Designed specifically for **incremental/lifelong object recognition**
- Includes task sequences for incremental learning benchmarks ([Lifelong Robotic Vision](#))

Links

- **Dataset page & downloads:** https://lifelong-robotic-vision.github.io/dataset/Data_Object-Recognition.html ([Lifelong Robotic Vision](#))
<https://www.kaggle.com/datasets/zhedamai/openlorisobject/discussion?sort=undefined>
- **Code repository & scripts:** <https://github.com/lifelong-robotic-vision/OpenLORIS-Object> ([GitHub](#))

2. OpenLORIS-Scene (Lifelong SLAM and Scene Understanding)

Overview

OpenLORIS-Scene provides **multi-sensor sequences** captured by a wheeled robot in real indoor scenes. It includes visual and inertial sensor data collected at different times to reflect **environment changes** caused by human activity, illumination, and object movement. ([Lifelong Robotic Vision](#))

Contents

- Color + depth images (RealSense D435i)
- Stereo fisheye images (RealSense T265)
- IMU measurements (accelerometer + gyro)
- Wheel odometry data
- Ground-truth trajectories via motion capture or LiDAR SLAM
- Multiple sequences per scene for long-term change evaluation ([Lifelong Robotic Vision](#))

Why It's Useful

- Real robot dataset for **lifelong SLAM**
- Supports incremental evaluation (sequence order matters)
- Includes sensor metadata and calibration information ([Lifelong Robotic Vision](#))

Links

- **Dataset page & metadata:** <https://lifelong-robotic-vision.github.io/dataset/scene.html> ([Lifelong Robotic Vision](#))
<https://huggingface.co/datasets/shixuesong/openloris-scene/tree/main/package>
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3. OpenLORIS-Location (Indoor Visual Localization)

Overview

OpenLORIS-Location focuses on visual localization within indoor spaces, captured by robots over different times and conditions. Images describe the **same real world location** under varying lighting and dynamics, useful for incremental/long-term localization tasks. ([Lifelong Robotic Vision](#))

Contents

- ~1553 RGB images across 93 locations
- Multiple images per location with appearance changes
- Indoor scenes: office, home, corridor, market, restaurant, station
- Conditions include illumination changes and dynamic object presence ([Lifelong Robotic Vision](#))

Why It's Useful

- Real world location changes over time

- Useful for incremental visual localization / mapping research ([Lifelong Robotic Vision](#))

Links

- **Dataset page & metadata:** <https://lifelong-robotic-vision.github.io/dataset/location.html> ([Lifelong Robotic Vision](#))
<https://github.com/lifelong-robotic-vision/OpenLORIS-Location/releases>
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4. LIBERO (Lifelong Robot Learning Benchmarks)

Overview

LIBERO is a **robot manipulation benchmark suite** designed explicitly for lifelong and incremental skill learning research. It includes multiple versions with varying numbers of tasks, emphasizing **generalization and skill transfer** across tasks. (lifelong-robot-learning.github.io)

Contents

- **libero_spatial:** spatial reasoning tasks
- **libero_object:** object manipulation tasks
- **libero_goal:** goal-oriented tasks
- **libero_100:** 100 diverse manipulation tasks
- Tasks represent incremental sequences of manipulation challenges (lifelong-robot-learning.github.io)

Why It's Useful

- Structured for **incremental skill learning evaluation**
- Growing number of tasks allows testing scalability and transfer (lifelong-robot-learning.github.io)

Links

- **Dataset documentation & downloads:**
https://lifelong-robot-learning.github.io/LIBERO/html/algo_data/datasets.html (lifelong-robot-learning.github.io) <https://huggingface.co/datasets/yifengzhu-hf/LIBERO-datasets>
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5. Open X-Embodiment (Cross-Robot Learning Dataset)

Overview

Open X-Embodiment is a **large unified robotics dataset** designed to support **multi-embodiment learning and skill transfer** across a variety of robot types (arms, humanoids, quadrupeds) with real data. ([GitHub](#))

Contents & Metadata

- Robot episodes captured in RLDS (Robotics Learning Data Storage) format
- **Metadata captures:**
 - camera intrinsics/extrinsics
 - robot kinematics/limits
 - task descriptors / skill IDs
 - success/failure labels
 - scene identifiers and contextual information
- Includes hundreds of thousands of episodes for cross-task and cross-robot learning ([GitHub](#))

Why It's Useful

- **Multi-robot, multi-skill real data**
- Ideal for **embodied learning and transfer**
- Standardized metadata amenable to incremental learning pipelines ([GitHub](#))

Links

- **GitHub & metadata references:** https://github.com/google-deepmind/open_x_embodiment ([GitHub](#))
- **Project overview & metadata info:** <https://robotics-transformer-x.github.io> (project website referenced on dataset pages) ([Emergent Mind](#))

How to Use the Metadata

For each dataset, **metadata is available either on the dataset pages or through accompanying GitHub repositories:**

Dataset	Where to Find Metadata
OpenLORIS-Object	Dataset page & GitHub repo (labels, difficulty, tasks) (Lifelong Robotic Vision)
OpenLORIS-Scene	Dataset page (sensor calibration, ground-truth, timestamps) (Lifelong Robotic Vision)
OpenLORIS-Location	Page with scene and location groupings (Lifelong Robotic Vision)
LIBERO	Documentation lists task splits, task metadata (lifelong-robot-learning.github.io)
Open X-Embodiment	GitHub and spreadsheet with metadata fields (robot type, skill labels) (GitHub)

References for Dataset Metadata

- OpenLORIS main page with links to all dataset variants and papers. ([Lifelong Robotic Vision](#))
- Detailed descriptions of OpenLORIS-Object environmental factors for incremental learning. ([Lifelong Robotic Vision](#))
- Sensor modalities and ground truth in OpenLORIS-Scene. ([Lifelong Robotic Vision](#))
- LIBERO dataset listings and download instructions. (lifelong-robot-learning.github.io)
- Open X-Embodiment dataset structure and metadata formats. ([DeepWiki](#))

- travailler sur l'aspect collaboratif avec ces datasets

1. HARMONIC — Human & Robot Multimodal Collaborative Dataset

Description:

A multimodal dataset of **human interactions with a 6-DOF robot arm** during assistive eating tasks. It includes:

- EMG (forearm) → muscle activation
- Eye gaze + egocentric and third-person video
- Robot joint positions
- Joystick commands
- Body pose, hand pose, facial keypoints

This is *research data* collected for shared autonomy and intent prediction. ([arXiv](#))

Metadata & Facts

- Subjects: 24 participants
- Modalities: video (multiple views), EMG, gaze, robot actions
- Formats: CSV, YAML, videos

Dataset / Metadata:

ArXiv Paper (with data description):

<https://arxiv.org/abs/1807.11154> ([arXiv](#))

(Download links typically found via authors' project page in the paper or consortium pages — search for “HARMONIC dataset download” linked from the paper.)

2. MultiPhysio-HRC — Multimodal Human-Robot Collaboration Physiology Dataset

Description:

A large dataset capturing **physiological, behavioral, and audio data** in real human-robot collaboration scenarios. Modalities include:

- EEG (brain activity)
- ECG (heart)
- EMG (muscle)
- EDA (skin conductance)
- Respiration
- Voice & video

Designed for **human status estimation (stress/cognitive load)** during collaborative tasks with robots. ([arXiv](#))

Metadata & Facts

- Rich physiological sensor streams
- Baseline annotations (psychological labels)
- Good for affective computing + HRC learning

Dataset / Metadata:

Data described in the paper; many such datasets provide download links or contact for access:

<https://www.mdpi.com/2218-6581/14/12/184> (MDPI)

3. RoboMNIST — Multi-Modal Multi-Robot Activity Dataset

Description:

A dataset using **two Franka Emika robot arms** with data from:

- WiFi channel state information (CSI)
- Video
- Audio

The dataset captures robot activities in context and is multimodal, though not wearable in the classic IMU/EEG sense. It's research-oriented and stored in Figshare/GitHub. ([Nature](#))

Metadata & Facts

- Multimodal: WiFi, video, audio
- Robot activity recognition focus
- Documentation and example loaders on GitHub

Dataset / Metadata:

Figshare repository + GitHub:

<https://github.com/SiamiLab/RoboMNIST> ([Nature](#))

4. A Dataset of Standard and Abrupt Industrial Gestures Recorded Through MIMUs

Description:

The dataset introduced in this paper (often referred to as **DASIG**) is created to support research in **human–robot collaboration (HRC)** with a focus on **motion recognition, ergonomics, safety, and gesture analysis** in industrial contexts. It specifically includes **both standard industrial gestures and**

unexpected/abrupt movements that may occur due to human inattention or environmental triggers, which is critical for safe HRC.

Dataset Content

Sensor Modality

- **MIMUs (Magneto-Inertial Measurement Units)**

Each MIMU provides:

- 3-axis accelerometer data
- 3-axis gyroscope data
- (possibly) magnetometer data depending on device used

This multimodal inertial data tracks **human arm and body motion** during gesture execution, capturing both smooth and sudden movements imposed by alarms or hazards.

Gesture Types

The dataset comprises two main categories:

1. **Standard Industrial Movements**

- Typical gestures such as pick-and-place, reaching, lifting
- Representative of regular operations in manufacturing or assembly

2. **Abrupt / Unexpected Movements**

- Movements triggered by **visual or acoustic alarms**
- Represent sudden, atypical motion due to inattention, hazards, or safety responses
This is important for *real HRC safety research*, because robots must detect and adapt to these events reliably.

Dataset / Metadata:

<https://zenodo.org/records/17660014>

Best Way to Build a Wearable-Controlled Human–Robotic Arm Using Incremental Learning

1. Core Design Principle (Very Important)

Incremental learning should learn the *mapping* from human intent → robot skill, not raw motor commands.

Why?

- Wearable signals are noisy
- Humans vary over time
- Robot safety must be guaranteed

So the robot **does not directly imitate the wearable** — it *interprets* it.

2. System Architecture (Research-Grade & Safe)

```
Wearable Sensors (EMG / IMU / Gloves)
    ↓
Human Intent Encoder (Neural Network)
    ↓
Incremental Learning Module
(Intent → Skill / Parameters)
    ↓
Skill Library (Frozen + Expandable)
    ↓
Motion Planner (IK / Trajectory)
    ↓
Low-Level Control (Impedance / PID)
    ↓
Robotic Arm
```

This architecture is **critical** and widely accepted in HRI research.

3. Wearable Device: What to Use (Realistic Options)

Option A — EMG Armband (Best for Intent)

- Muscle activation signals
- Strong link to human intent
- Used in prosthetics & HRI

Examples:

- Myo armband

- Delsys EMG
- Custom EMG sensors

Option B — IMU-based Wearables (Best for Motion)

- Wrist orientation
- Arm motion
- Low cost and robust

Often combined with EMG.

Option C — Smart Gloves (Dexterous Control)

- Finger flexion
- Grasp intent
- Force estimation

Best Practice

Combine EMG + IMU

- EMG → *what* the human wants
- IMU → *how* the human moves

4. What Incremental Learning Learns (Key Decision)

You have **three good learning targets**, ordered from safest to most powerful:

Skill Selection (Best Starting Point)

Wearable → *Which skill to activate*

- Pick
- Place
- Rotate
- Hand-over

Incremental learning adds **new skills over time**.

Skill Parameter Adaptation

Wearable → *Parameters of a skill*

- Speed
- Grip force
- Direction

- Precision level

Incremental learning adapts to:

- New users
- Fatigue
- Context changes

Policy Refinement (Advanced)

Wearable → *high-level action policy*

- More flexible
- Needs safety constraints

5. Incremental Learning Algorithms (Best Choices)

Replay + Distillation (DER / DER++) — Best Overall

Why it's ideal here:

- Wearable data drifts over time
- Human behavior changes
- Keeps old skills stable

How it works in your system:

- Store a small buffer of old wearable–skill pairs
- Store old network outputs
- Train on:
 - New wearable data
 - Old “remembered” behavior

Used in **robot manipulation and HRI research**.

Adapter Modules (Highly Scalable)

- One adapter per user or skill
- Shared backbone

Perfect for:

- Multi-user systems
- Long-term deployment

Avoid (for this use case)

- Pure EWC / SI → too weak for noisy wearable data
- Progressive NNs → too heavy for real robots

6. Human Teaching Loop (Incremental Learning in Practice)

Step-by-step:

1. Human wears device
2. Demonstrates or corrects motion
3. Robot executes a safe version
4. Human gives:
 - Confirmation
 - Correction
5. Data stored incrementally
6. Model updated *without forgetting*

This creates a **human-in-the-loop lifelong learner**.

7. Dataset Strategy (Realistic & Publishable)

Initial Dataset

- 5–10 skills
- 2–3 users
- 10–20 demonstrations per skill

Record:

- Wearable signals
- Robot state
- Selected skill
- Outcome (success/failure)

Incremental Phase

Each update introduces:

- A new user **or**
- A new skill **or**
- A change in wearable placement

This is **perfect for evaluating incremental learning**.

8. Safety Layer (Mandatory)

Even if learning fails:

- Impedance control
- Joint limits
- Speed limits
- Human override

The learned model **never directly controls motors**.

9. Example Technology Stack

Component	Recommendation
Middleware	ROS 2
Wearable	EMG + IMU
Learning	PyTorch
IL Method	DER + replay buffer
Skills	DMPs or motion primitives
Planner	MoveIt
Controller	Impedance control

- download datasets

- **MultiPhysio-HRC — Multimodal Human-Robot Collaboration Physiology Dataset**

what did they do with it, papers...

how can we exploit it in our project